Cardiovascular System: Physiology and Diseases

Integrated Biomedical and Clinical Sciences OMFS 512

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COLLEGE of DENTAL MEDICINE

Mission: To advance the science and art of health through education, service, scholarship and social accountability

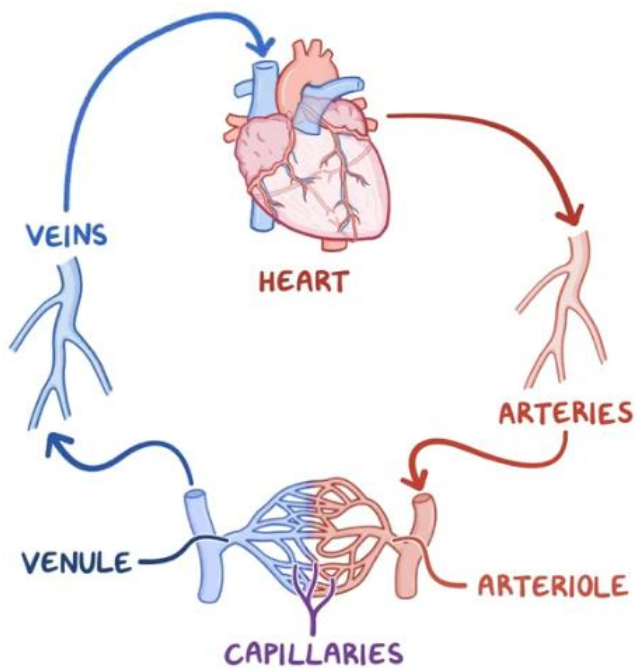
Learning Objectives

1. Describe the structure and function of the cardiovascular system.
2. Compare and contrast the systemic and pulmonary circulation.
3. Describe the electrophysiology of cardiac contraction.
4. Describe the cardiac output and its regulators.
5. Describe different categories of cardiovascular diseases and congenital heart conditions.
6. Describe different types of blood vessels and blood pressure regulation.

Outline

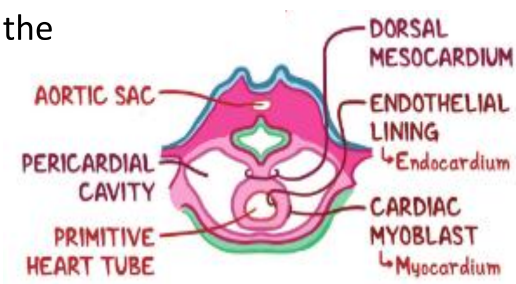
- Overview of the cardiovascular system
- Cardiac electrophysiology
- Determinants of cardia output
- Cardiovascular diseases
- Regulation of blood pressure
- Vascular diseases

Cardiovascular system

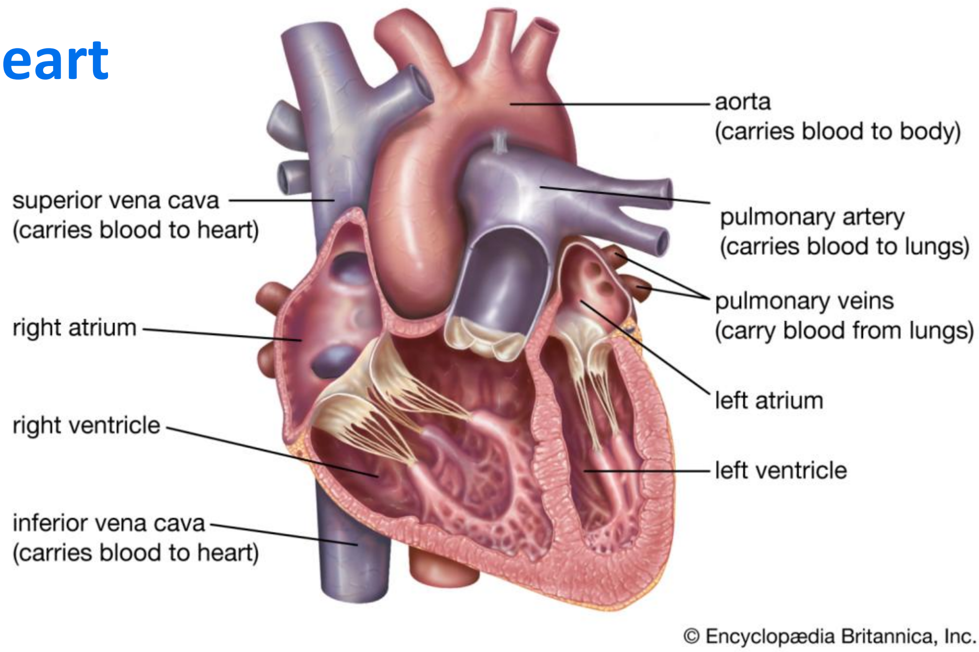


Embryonic development

- Begins in week 3 from the mesoderm layer.
- By the end of week 4, the heart is pushed to the thorax and undergoes looping to generate 2 primitive atria and 2 primitive ventricles.
- Blood cells and blood vessels arise from the mesoderm.
- Blood vessels form through:
  - **Vasculogenesis**, blood vessels arise from blood islands.
  - **Angiogenesis**, blood vessels sprout from existing vessels.



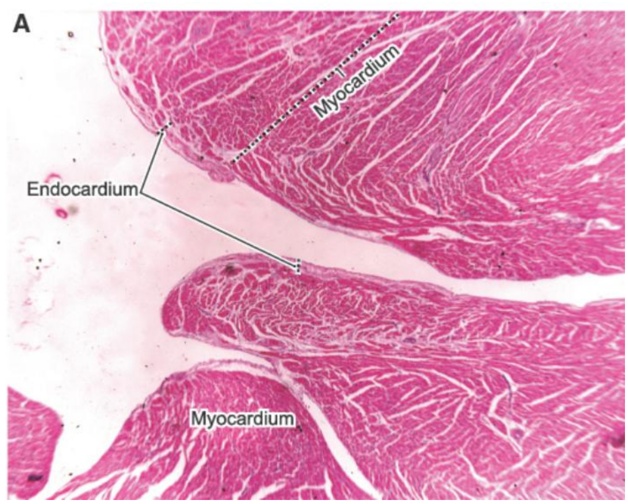
The heart


aorta
(carries blood to body)
superior vena cava
(carries blood to heart)
pulmonary artery
(carries blood to lungs)
pulmonary veins
(carry blood from lungs)
right atrium
left atrium
right ventricle
left ventricle
inferior vena cava
(carries blood to heart)
© Encyclopædia Britannica, Inc.

Histology

1-Endocardium (innermost layer).
Lines the lumen of the heart.
Made up of endothelium, subendothelium (thin loose connective tissue) and subendocardium.

Purkinje fibers are in the subendocardium.

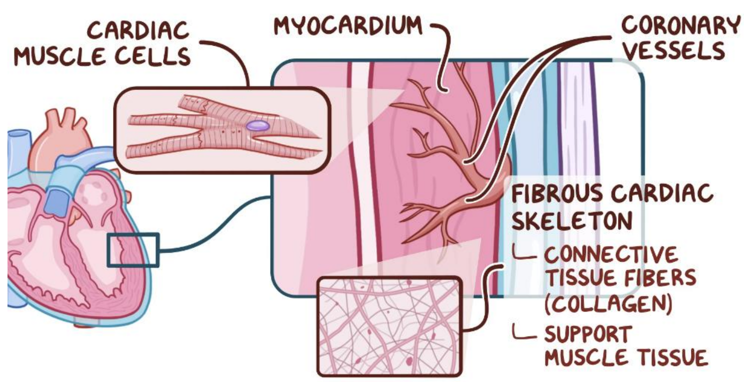


2- Myocardium (cardiac muscle).

Contractile cardiac muscle cells & collagen fibers.

Contracts to propel blood for distribution

Coronary arteritis supply oxygenated blood to the myocardium.

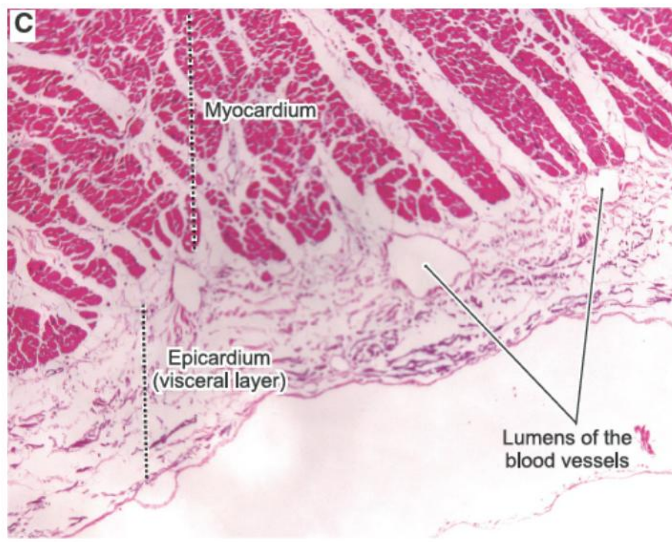


3- Epicardium

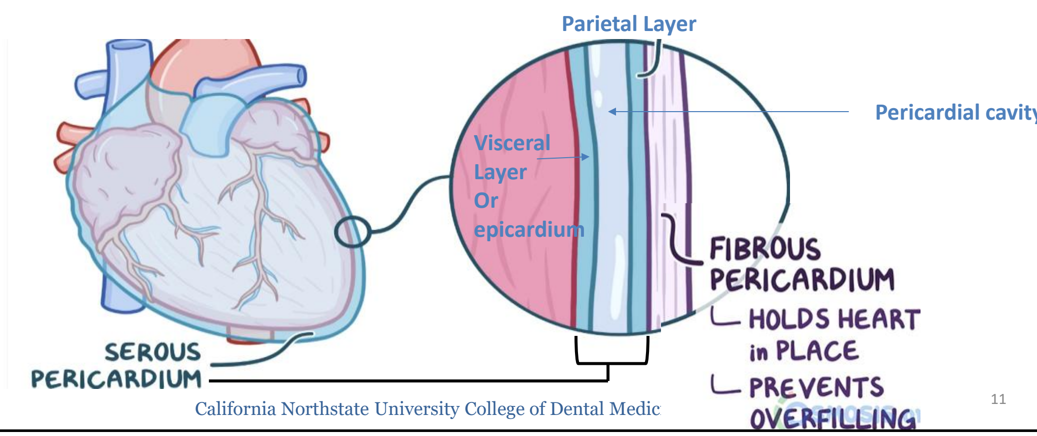
(outermost layer, aka, viscera pericardium).

Thin layer of connective tissue that houses roots of coronary vessels, nerves and adipocytes.

Constitutes the visceral part of pericardium. Functions to protect the heart. Role in embryonic heart development and injury response in role in myocardial infarctions.

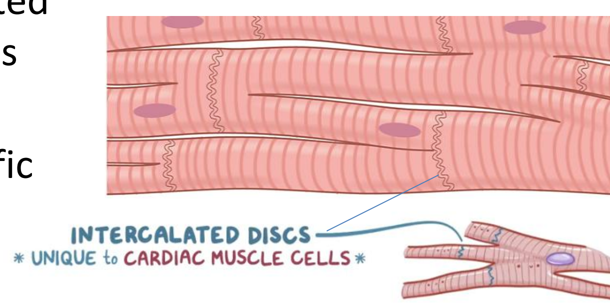


- The heart is enclosed in a **pericardium (fibrous pericardium + serous pericardium).** Separated from the epicardium by pericardial cavity.
- Serous layer secrete and reabsorb lubricant protein-rich fluid of the pericardial cavity, serves as lubricant and cushions heart during contraction.



Cardiomyocyte structure

- Cardiomyocytes are branched striated muscle fibers with intercalated discs exist at the ends of the cells.
- Intercalated discs are cardiac-specific structure (desmosomes+ gap junctions) to allow synchronized cardiac contractions

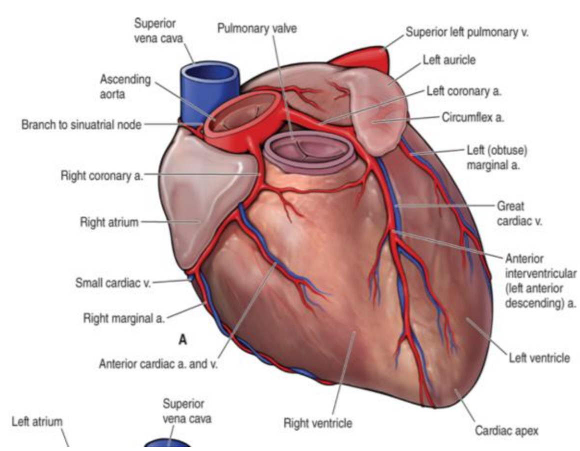


Calcium induces cardiomyocyte contraction via facilitating actin and myosin binding.

Coronary Circulation

- **Arterial supply:** Right & left coronary arteries (RCA &LCA) and their branches supply oxygenated blood. Subendocardium of left ventricle is the most common site of myocardial infarction and ischemic injury.

- **Venous return:** Most cardiac veins drain into coronary sinus which drain directly into the right atrium.

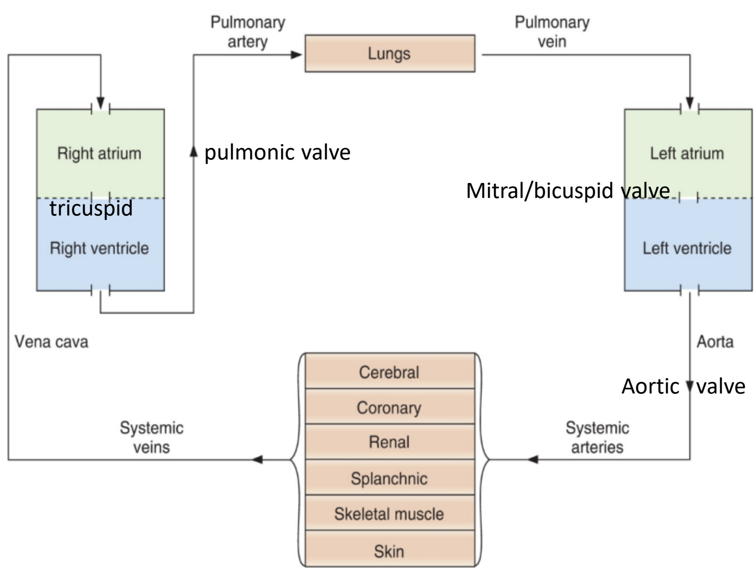


The heart as a pump !
Electrophysiology of the heart contraction

Cardiovascular Circulation

- **Pulmonary circulation**
- **Systemic circulation**

Difference in function & pressure



[VitalSource Bookshelf: BRS Physiology]

Cardiac cycle

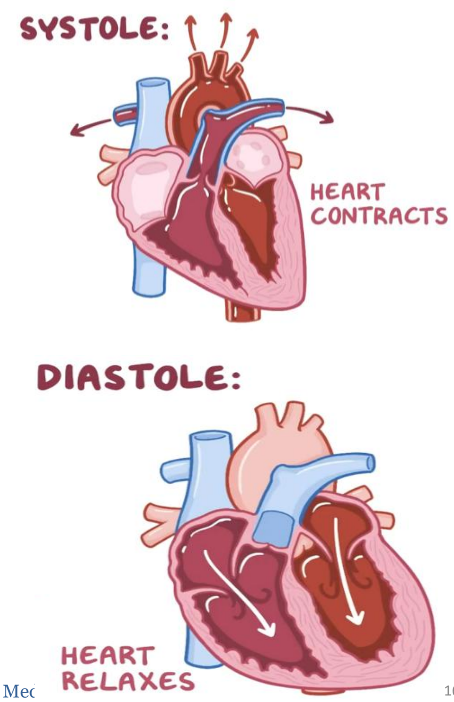
Mechanical and electrical events that occur in one heartbeat.

1- Systole phase:

Heart contraction phase. Ventricles pump the blood out. Bicuspid and tricuspid closed.

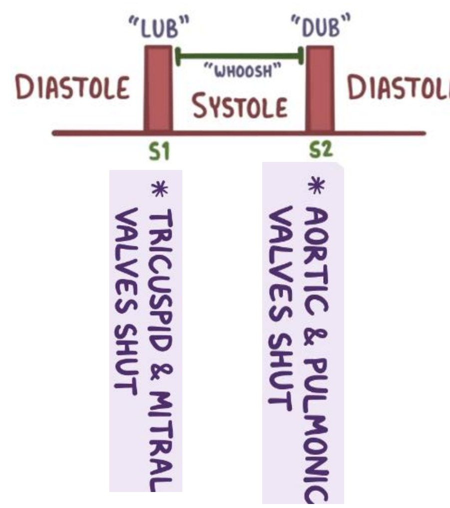
2- Diastole phase:

Relaxation of the heart muscle. During this phase, heart chambers are being filled.



Heart Sounds

- Mechanical event during cardiac cycle yield heart sounds.
  - S1 “LUBB” = auriculoventricular valve closing & sound projected onto the cardiac wall “soft sound”
  - S2“DUB”= closure of semilunar valves “sharp”
- Variations in heart sounds can indicate volume overload, structural abnormalities or muscle stiffness.



[The patient with a heart murmur - The Journal of the American Dental Association]

Murmur

- Heart murmur= abnormal heart sound due to turbulent blood flow through the heart.
- Can be developmental (normal) or indicate heart pathology (congenital heart conditions, Afib or valve disease
  - Valve regurgitation = abnormal valve closure = backflow of blood.
  - Valve stenosis= valve narrowing = hinder efficient blood flow.
- Consult PCP for causes and assess the need for ab prophylaxis

Cardiac Conduction

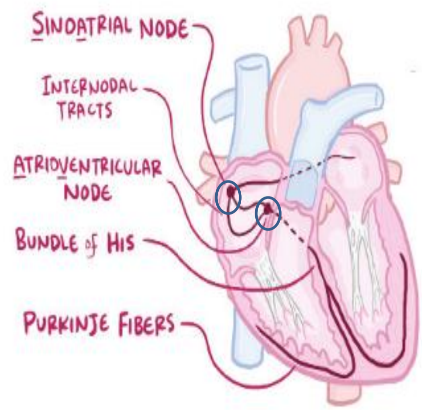
According to the electrophysiological function cardiomyocytes classified into:

1- **Slow depolarizing** or pacemaker cells

Clustered in the sinoatrial (SA) node and atrio-ventricular (AV) node. Constitute 1 % of the heart cells. Calcium channels.

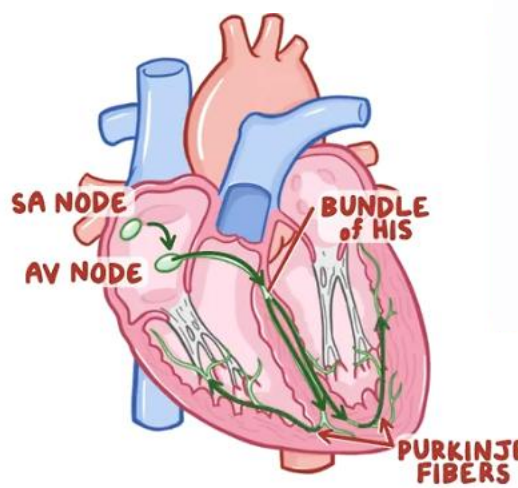
2- **Fast depolarizing cells** (work myocytes)

Atrial and ventricular myocardial cells, and cells of the His–Purkinje system. They represent 99 % of the heart cells. Fast sodium ion channels.



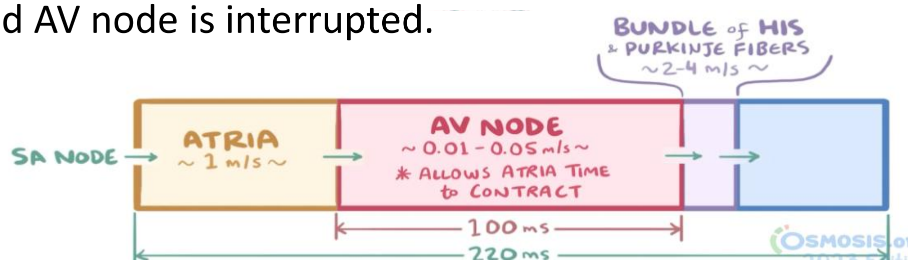
Cardiac Conduction (*cont.*)

- The pacemaker cells in the SA node generate spontaneous action potential that starts sequential chain of events leading to the cardiac contraction.

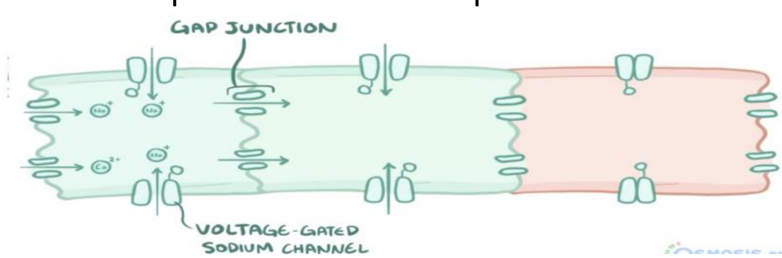


SA node → Atrial contraction → AV node → ventricular bundles (His/Purkinje) → ventricular contraction.

- This whole process takes around 200 millisecond.
- At the AV node, conduction is delayed by approximately 100 ms to allow completion of atrial contraction (= blood ejection from the atria and complete ventricular filling).
- The AV node also serves as a backup pacemaker in situations when the SA node ceases to function or communication between the SA node and AV node is interrupted.



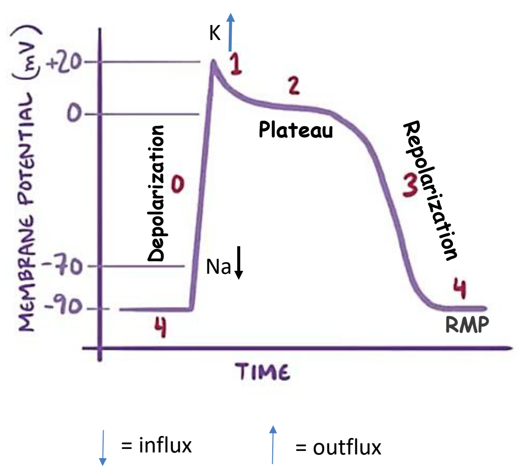
- Action potentials or depolarization waves are rapidly transferred across the neighboring cardiomyocytes via gap junctions. This process triggers the cardiac contraction.
- Ventricles, atria, and the Purkinje system have resting membrane potentials of about −90 mV. Equal to the K^+ equilibrium potential
- Sodium and calcium ions can not move freely across the cell membrane due to the membrane selective permeability.
- Depolarization = more positive membrane potential.



Transfer of the action potential in cardiomyocytes

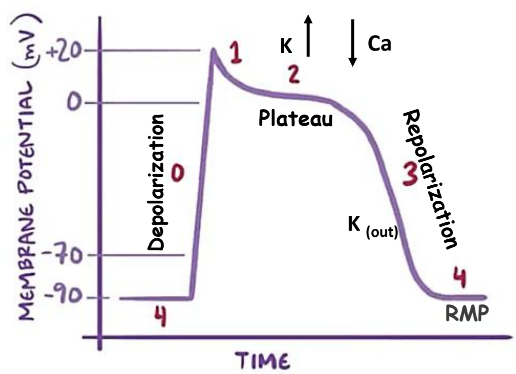
1- Phase 0, depolarization phase: Action potential from pacemaker = open sodium channels.

2- Phase 1: At threshold of +20 millivolt, sodium channels are closed, and potassium channels are open. Potassium slowly moves out of the cells changing membrane potential to +10 millivolts "repolarization starts".



3- **Phase 2**, plateau (at +10 mV)
Voltage-gated calcium channels are open, calcium influx and balances potassium outflux. Calcium influx induces calcium release from internal calcium stores and triggers the cardiomyocyte contraction.

4- **Phase 3**, rapid repolarization phase: following contraction, calcium channels (closed), and potassium outflux regain the resting membrane potential of -90 millivolts.

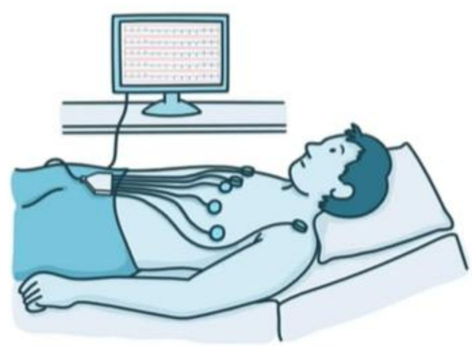


RMP= resting membrane potential

5- **Phase 4,** the resting membrane potential of -90 millivolts is regained. Potassium channels are open and potassium ion freely moving across the membrane.

Electrocardiogram

- Electrocardiogram (ECG) records the electrical heart activity using electrodes placed on the chest and arms.
- Used to monitor the cardiac conduction cycle and cardiac function.
- Clinical application: Diagnosing arrhythmia, MI, structural heart changes.



P wave= Atrial depolarization.

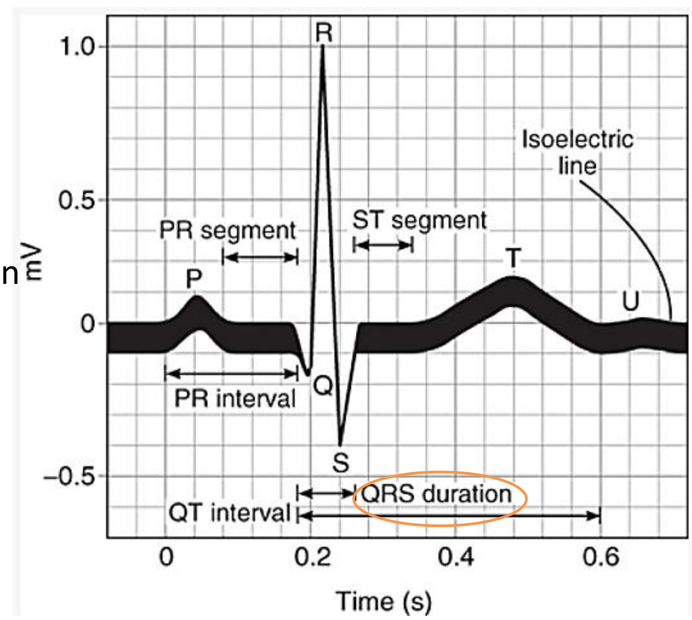
PR interval = time for atrioventricular conduction

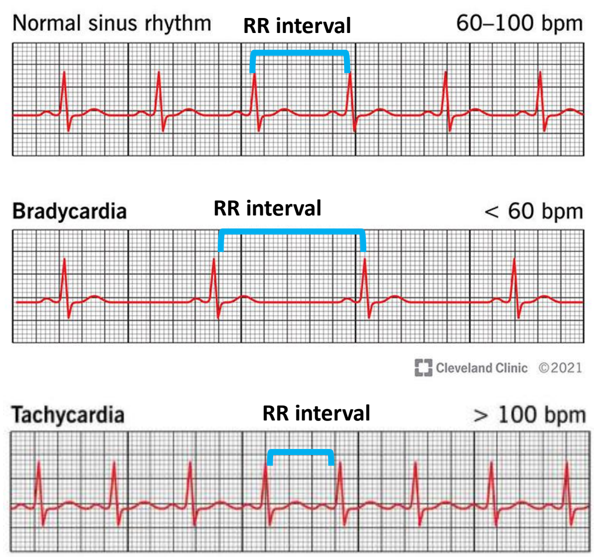
QRS = Ventricular depolarization

T wave = Ventricular re-polarization

Macrolide ab. & other drugs can prolong the QT interval leading to arrythmia as an adverse effect.

QT represent ventricular systole

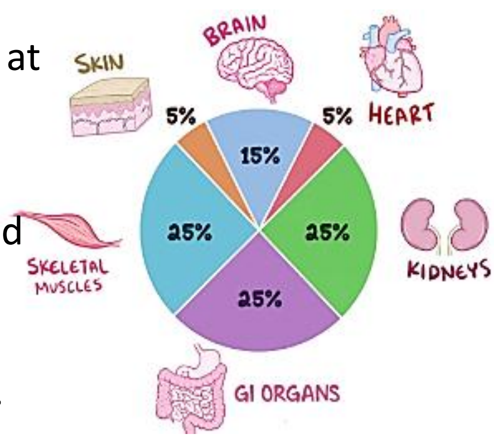



Normal sinus rhythm
RR interval
60–100 bpm
Bradycardia
RR interval
< 60 bpm
Cleveland Clinic ©2021
Tachycardia
RR interval
> 100 bpm

Determinants of Cardiac Function

Cardiac Output

- Cardiac output (CO) = amount of blood pumped by the heart per min (5-6 L/min at rest).
- Mean value in a 70-kg male = 5 L/min.
- CO is a function of the heart rate (HR) and stroke volume (SV). CO = HR x SV.
  - HR is the number of beats per minute.
  - SV is the volume of blood pumped per beat.



- Under physiological conditions, the cardiac output is adjusted according to metabolic demands (at rest vs during exercise).
  - At rest, CO is around 5 L/min (70 bpm x 70 ml/beat)
  - During exercise, CO can reach 20 -35 L/min
- CVS is a closed system. Thus, venous return to both atria is --------------- to the CO. CO of the right ventricle ---------- CO of the left ventricle.
- CO is altered in cardiovascular pathologies.

Determinants of the cardiac output

1- Heart rate

Rate of heartbeat. It is regulated by the autonomic nervous system. Normal 60 -100 bpm.

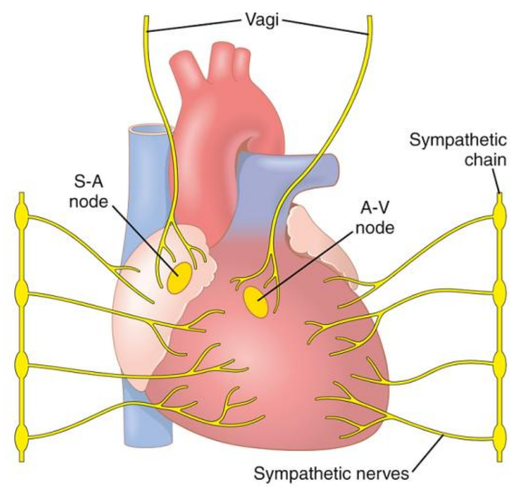
2- Stroke volume

Volume of blood pumped per heartbeat. It is affected by the sympathetic nerve activity, the arterial and ventricular-filling pressure and depends on:

A. Preload
B. Afterload
C. Contractility of the cardiac muscle

Autonomic control

- **PSNS:** vagal innervation for the SA node, atria, and AV node but not the ventricles. At rest, the parasympathetic drive is responsible for reducing the frequency of action potentials to yield normal resting HR
- **SNS:** innervates the heart β1 receptors. can increase HR in young adult from 70 beats/min up to 180 to 200 beats/min



Impact	Sympathetic regulation	Parasympathetic regulation
Neurotransmitter	Norepinephrine	Acetylcholine
Heart rate “chronotropic effect”	Increase the heart rate via acting on β receptor (positive chronotropic effect)	Decrease the heart rate via muscarinic receptors
Contraction “inotropic effect”	Increase the contraction strength via acting on β receptor	Decrease the strength of atrial contractions via activating the muscarinic receptors
Conduction velocity “dromotropic effect”	Increases conduction velocity through the AV node and decreases the PR interval.	Decreases conduction velocity through the AV node and increases the PR interval.

Determinants of Stroke Volume

A. Preload: defined by the degree of myocardial stretch before contraction. The more the muscle will stretch the more blood can be pumped out.

Preload determinants:

- Preload is enhanced by increased amount of blood returned to the ventricles "venous tone".
- Preload is decreased in hemorrhage and dehydration.

B. Afterload: defined by the resistance that the ventricles need to overcome to pump the blood out. The higher the arterial and vascular resistance, the higher the afterload & lower stroke volume.

Afterload determinants:

Increased arterial pressure decreases stroke volume by increasing the afterload on cardiac muscle fibers. Therefore, afterload is affected by hypertension, systolic heart failure.

C. Contractility: Natural ability of the myocardium to contract independent of preload or afterload. Directly proportional to stroke volume.

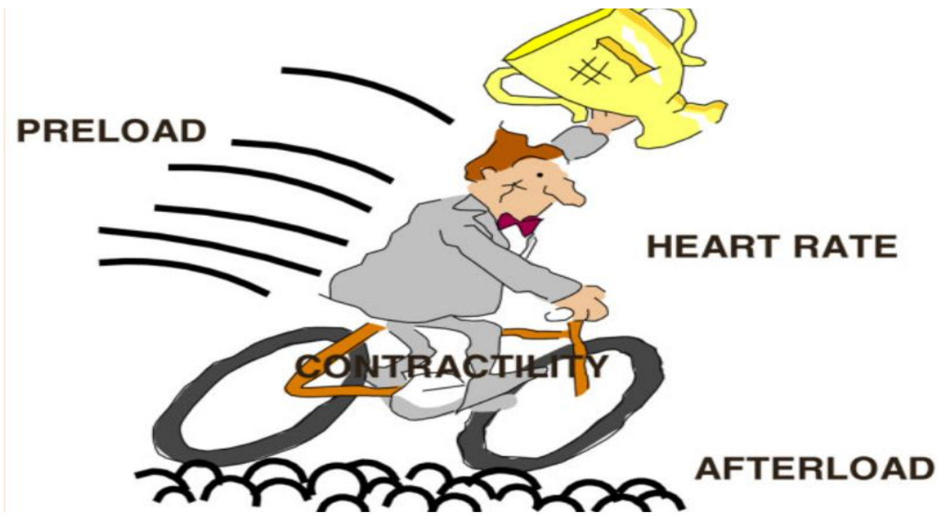
Contractility determinants:

I) Factors that increase contractility:

1- Norepinephrine
2- Inotropic drugs (like digoxin and dobutamine).

II) Factors that decrease contractility :

1- Acetylcholine
2- Beta blockers or calcium channel blockers
3- Hypoxia: decrease the available energy for cardiomyocytes.



Contractility and Preload are directly proportional to the stroke volume while the afterload is inversely proportional to stroke volume.

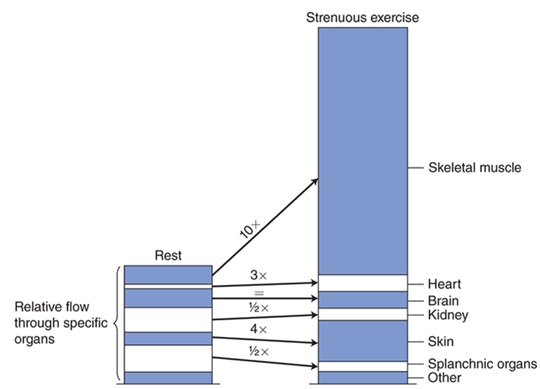
Understanding cardiac output - PMC (nih.gov)

Ejection Fraction

- Ejection fraction reflects the efficiency of myocardial contractility.
- EF = the volume of blood pumped from the left ventricle per beat (stroke volume) **/** the volume of blood in the left ventricle at the end of diastole (contraction) x100
- Normal EF is around 55 to 60%.
- EF< 50 % indicates low cardiac contractility

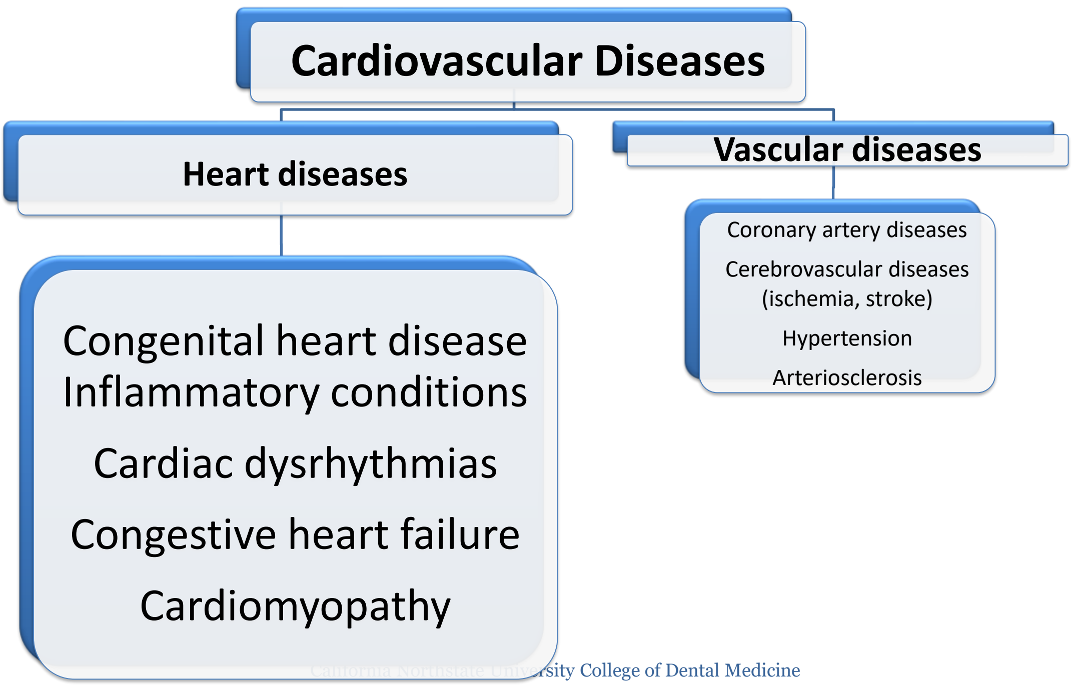
Physiological Response to Exercise

- Muscle contraction= ATP consumption and low oxygen levels.
- Local hypoxia lead to vasodilation which activate SNS.
- SNS induce tachycardia & increase CO = more blood to muscles.
- Also, SNS increase sweating to decrease core body temp.



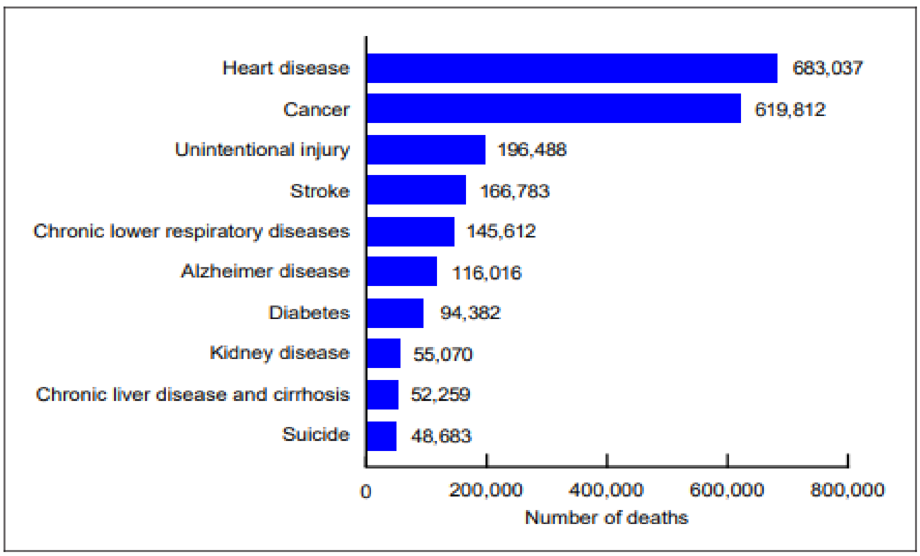
	Rest	Strenuous exercise
Cardiac output	6 L/min	18 L/min
Heart rate	70 beats/min	160 beats/min
Ejection fraction	60%	80%
Arterial pressure	120/80 mm Hg	150/80 mm Hg
Central venous pressure	2 mm Hg	2 mm Hg

Source: David M. Harris: *Mohrman and Heller's Cardiovascular Physiology, 10th Edition*


Cardiovascular Diseases
Heart diseases
Vascular diseases
Congenital heart disease
Inflammatory conditions
Cardiac dysrhythmias
Congestive heart failure
Cardiomyopathy
Coronary artery diseases
Cerebrovascular diseases (ischemia, stroke)
Hypertension
Arteriosclerosis

Heart diseases are still a leading cause of death

Figure. Leading underlying causes of death: United States, 2024



NOTES: Deaths that occurred in the United States among residents of U.S. territories and foreign countries were excluded. Estimates for 2024 are based on provisional data as of June 1, 2025 (provisional mortality data are available from https://wonder.cdc.gov). Provisional data are incomplete, and data from December are less complete because of reporting lags.
SOURCE: National Center for Health Statistics, National Vital Statistics System, mortality.

NCHS Provisional Report on U.S. Mortality for 2024 | Blogs | CDC

Risk factors

Modifiable risk factors	Unmodifiable risk factors
Excessive consumption of saturated fats, salts	Older age (specially for arteriosclerosis)
Smoking	Male gender
Sedentary lifestyle and Obesity	Genetic predisposition
Diabetes	Ethnicity
Stress	
Dyslipidemia (elevated serum lipids)	
High blood pressure	

Diagnosis of Cardiovascular Diseases

- **Clinical manifestation**
- **Laboratory investigation** to detect enzyme leakage that can reflect cardiomyocyte damage.
  - ➢ Myoglobin, **cardiac-specific troponins** and creatine kinase-MB fraction are sensitive biomarker for myocardial damage.
  - ➢ Cholesterol and serum lipids (LDL/HDL)
- **Imaging studies:** electrocardiogram "ECG" to detect abnormalities in the cardiac rhythm; 3D CT scans and/or MRI that are usually used to detect structural changes of the heart.

Congenital heart diseases

Congenital heart diseases (CHD)

Structural abnormalities of the heart or great vessels that occur during fetal development and presents at and soon after birth. Categorized into:

1. **Acyanotic:** don't restrict the amount of oxygenated blood to the body. Examples include VSD and ASD.

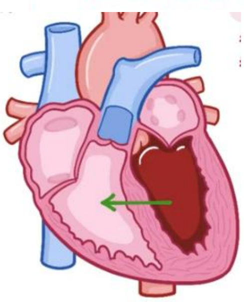
2. **Cyanotic:** heart defects that reduces the amount of oxygen delivered to the body. Examples include tetralogy of Fallot and transposition of the great vessels.

LEFT-TO-RIGHT SHUNT
* CALLED ACYANOTIC (NO CYANOSIS)

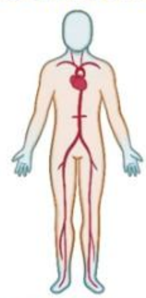
HIGH YIELD

* ASYMPTOMATIC
* SIGNS of HEART FAILURE
  - EXERCISE INTOLERANCE
  - SHORTNESS of BREATH

INFANTS / YOUNG CHILDREN
  - POOR FEEDING
  - FAILURE to THRIVE

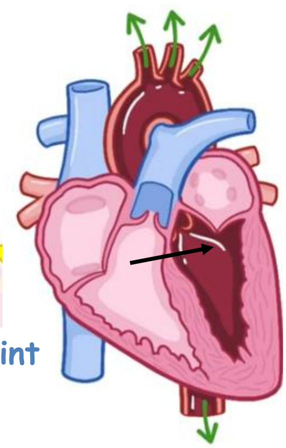


RIGHT-TO-LEFT SHUNT
* DEOXYGENATED BLOOD in SYSTEMIC CIRCULATION
  ↳ CYANOSIS



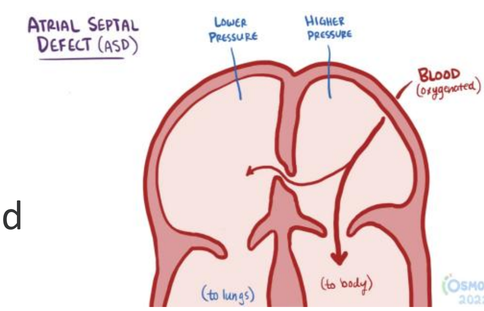
* CYANOTIC HEART DISEASES

Skin has a bluish tint



1a- Atrial septal defect (ASD)

Etiology: Defect in the middle or lower part of atrial septum. It cause the oxygenated blood to get into the pulmonary circulation.

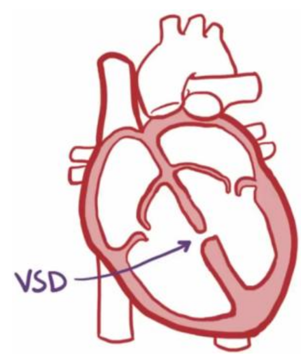


Left-right shunt

Clinical presentation: Small defects are asymptomatic. Large defects are manifested by excessive fatigue, murmur, difficulty breathing and inability to thrive.

1b- Ventricular septal defect (VSD)

Etiology: Defect in the muscular or membranous portion of the ventricular septum. Most common defect. It causes the flow of oxygenated blood from the left to right ventricles (left to right shunt).



Left-right shunt

Clinical presentation: fatigue, fast breathing, heart murmur. Most VSDs are detected and closed/improved during childhood.

2a- Tetralogy of Fallot

Most common CCHD. Usually with Down's and DeGeorge syndromes.

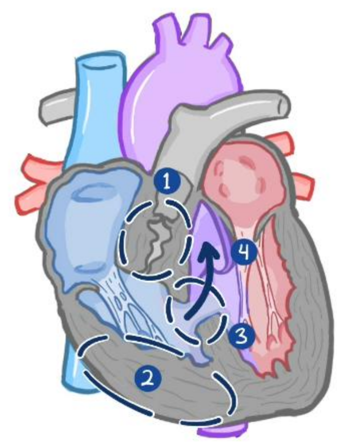
Etiology: Four heart abnormalities:

1- Stenosis of the ventricular tract to the pulmonary artery (hinders deoxygenated blood flow to lungs).

2- Right ventricular hypertrophy.

3- VSD leading to right to left shunt.

4- Aortic override for the septal defect (i.e, present in abnormal location).



Right-left shunt

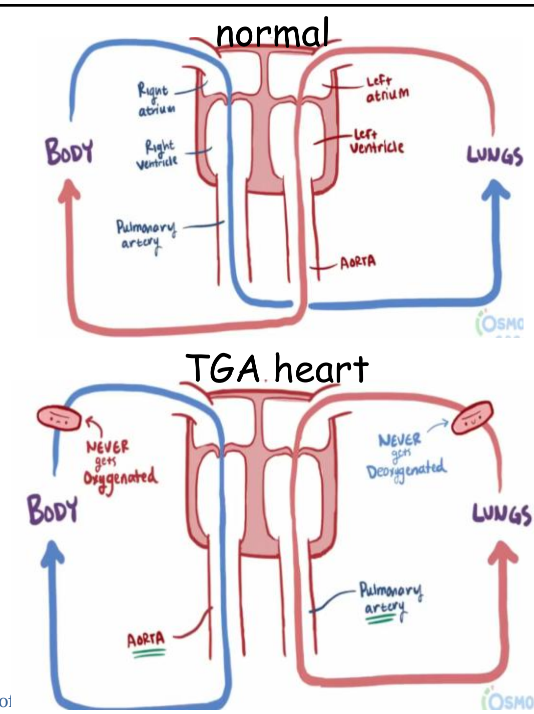
Clinical presentation: Progressive hypoxemia in childhood and heart murmur, among others.

2b- Transposition of the great vessels

"Switching the position of arteries"

Etiology: Congenital defect leading to the abnormal rearrangement of the great vessels (aorta with pulmonary artery). Second most common cyanotic congenital conditions.

Clinical presentation: Cyanotic newborn due to imbalance of oxygen level. Quick surgical intervention or can lead to death.



Dental Implications of CHD

- Children with CHD are at increased risk of infective endocarditis with higher mortality rate (compared to normal population).
- In some cases, children with CHD have more enamel defects (vs children without CHD). CHD can lead to increased risk of caries due to poor oxygenation or medication side effects.
- Children with unrepaired defects will have lower tolerance to handle stress, lengthy appointments ,…..
- Consultation and working with health care team. Antibiotic prophylaxis is recommended for unrepaired cyanotic CHD and repaired CHD with prosthetic device (first 6 months post-op or w residual defects).

The dental management of children with congenital heart disease following the publication of Paediatric Congenital Heart Disease Standards and Specifications | British Dental Journal

Inflammatory heart disease

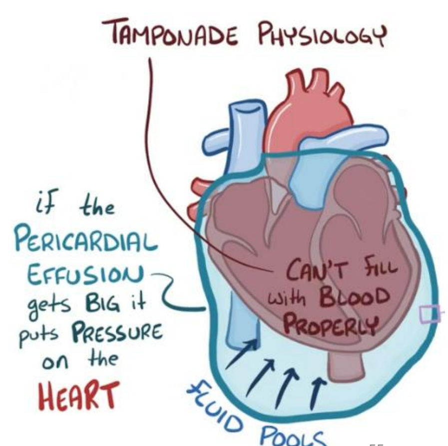
Pericarditis

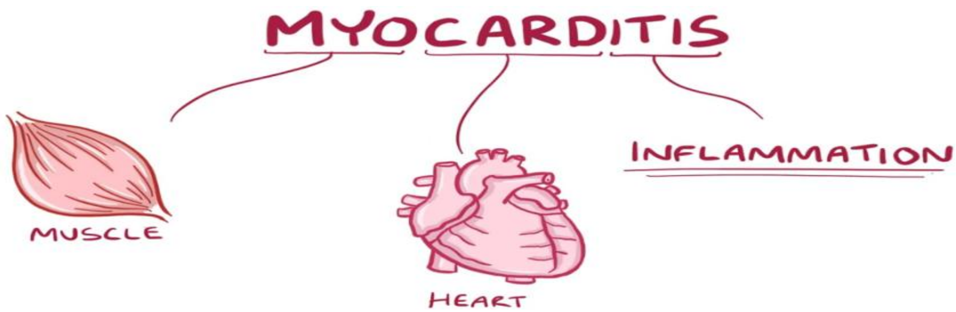
INFLAMMATION in the PERICARDIUM

- Inflammation and swelling of the pericardium that can lead to pericardial effusion “fluid accumulation in the pericardial sac”.
- **Causes**: idiopathic (common in acute pericarditis) or secondary to viral infections (coxsackievirus), after a myocardial infarction (secondary to apoptosis & inflammation) or in autoimmune disorders like lupus or trauma.
- **Symptoms** include fever and chest pain that worsens with deep breathing but improves with leaning forward.

- Pericarditis can increase the risk of pericardial effusion when inflammation causes fluid to accumulate around the heart. In worsening situations of pericardial effusion, a condition known as cardiac tamponade can develop.

- **Cardiac tamponade** is a compression of the heart muscle due to large fluid accumulation in pericardial cavity (why?)
- Dyspnea (shortness of breath) is a most common symptom of pericardial tamponade. It can lead to a medical emergency
- Treatment targets the source of inflammation and sometimes pericardiocentesis to drain the excess fluid.





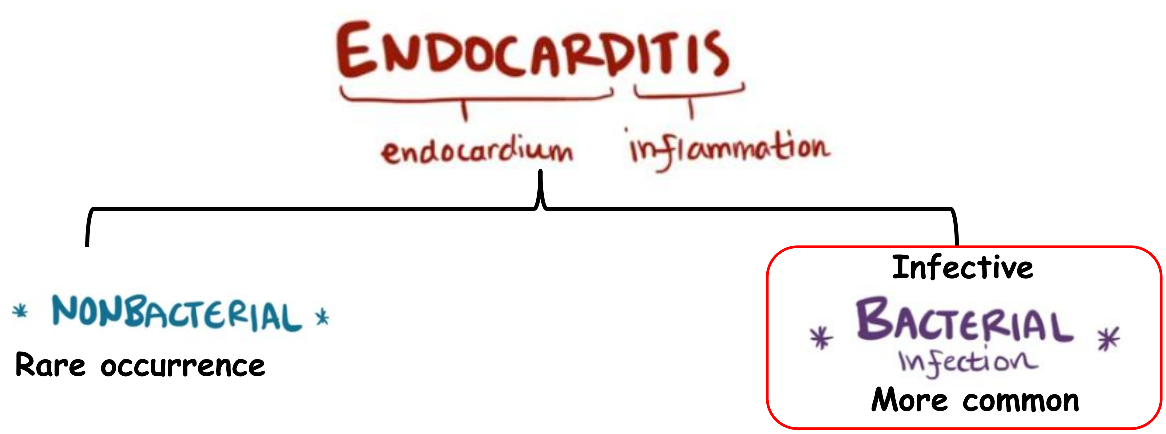
Causes:

1-Infection: viral (*adenovirus, coxsackievirus B)*, bacterial (*Streptococcus, Staphylococcus and Cyanobacterium diphtheria).*

2- Secondary to autoimmune conditions (systemic lupus erythematosus or rheumatoid arthritis).

3- Secondary to medications like sulfonamides or idiopathic.

Symptoms: Asymptomatic or characterized by chest pain, fever and arrythmia.



Infective endocarditis (IE) is a possibly life-threatening inflammation of the endocardium and the valves secondary to a bacterial infection

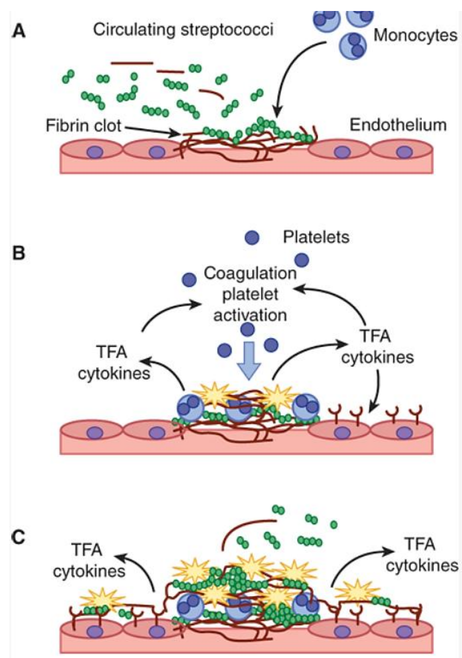
Infective endocarditis

- Caused by bacteremia following IV drug use, an abscess or oral flora after dental/gingival manipulation.
- Valvular heart disease. Most cases, infection involves left side heart (mitral and aortic valves). Drug users: infection of right heart side (tricuspid or pulmonary valve) may be seen.
- Risk factors include prosthetic valves, structurally abnormal cardiac valves (valvular disease; most common cause), previous rheumatic heart disease, congenital heart disease, hemodialysis and IV drug use.

Pathophysiology: damage to the endothelial lining of heart serves as a primary event that allows platelet aggregation and the formation of tiny clots.

Subsequent bacterial invasion (bacteremia) usually colonize those tiny clots and start an infection leading to vegetation or biofilm formation.

Sustained bacteremia stimulates immune system yielding sustained inflammation.



Source: Gary D. Hammer, Stephen J. McPhee: Pathophysiology of Disease: An Introduction to Clinical Medicine, Eighth Edition

Signs/Symptoms include fevers, chills, fatigue, chest pain, heart murmur & petechiae on extremities or mucous membranes.

Diagnosis: echocardiogram, imaging, clinical (murmur, fever, night sweats, risk factors), blood culture.

Management: antibiotic therapy (empiric & definitive for 4-6 weeks) and in some cases surgical valve replacement.

According to the cause, IE can be classified into:

Acute infective endocarditis: develop rapidly.

Affect normal heart valves and usually associated with *Staph aureus* (30 %). Associated with large vegetation that can damage the heart valve. Usually, introduced from the skin.

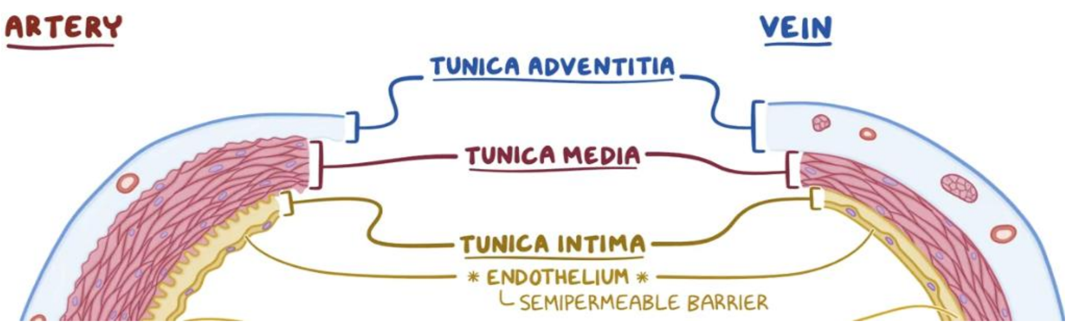
- Endocarditis in IV drug users: May involve infection by methicillin-resistant *S aureus* (MRSA).
- Early prosthetic heart endocarditis: Usually associated with skin flora, *Staph. epidermis or S. aureus,* within 60 days of valve replacement.

Subacute infective endocarditis: develop slowly in patients with predisposing valvular conditions like prosthetic valves.

- Characterized by small vegetation that don't damage the heart valve.
- Usually associated with bacteremia from oral surgery or gingival manipulation.
- Caused by viridians streptococci (oral flora, commonly *S. sanguis* and *S. mitis) or Enterococci endocarditis* (gut bacteria, elder population following GI surgery).
- Antibiotic prophylaxis is advised in patients with prosthetic valves or patients with prior history of IE.

Complications are more common among older patients, patients with S aureus infections, and those with prosthetic valve infections.

Vascular system



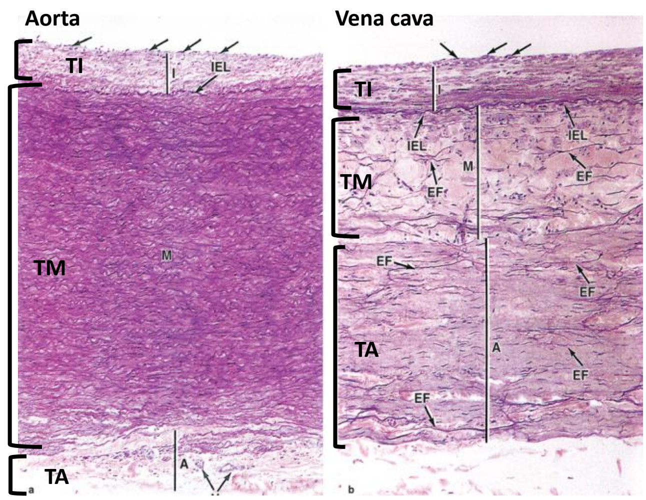
1- Tunica intima: Internal elastic layer; lined by endothelial cells that creates a low friction surface to allow smooth blood flow.
Endothelial cells synthesize nitric oxide (NO), a potent vasodilator, in response to a variety of chemical and physical stimuli.

Atherosclerosis is a common vascular disease that is characterized by the deposition of whitish-yellow plaques (atheroma) in the tunica intima.

2- Tunica media: Made up of smooth muscle, collagen or elastin fibers in concentric layers surrounding the lumen of blood vessels.

Size and components of the tunica media varies.

- Arteries have more elastic fibers compared to veins.
- Arterioles: mainly smooth muscle cells.



Arterioles resist the pressure created by the ventricular contraction and control the blood flow to capillaries (peripheral resistance). Therefore, arterioles are important for controlling the systemic blood pressure.

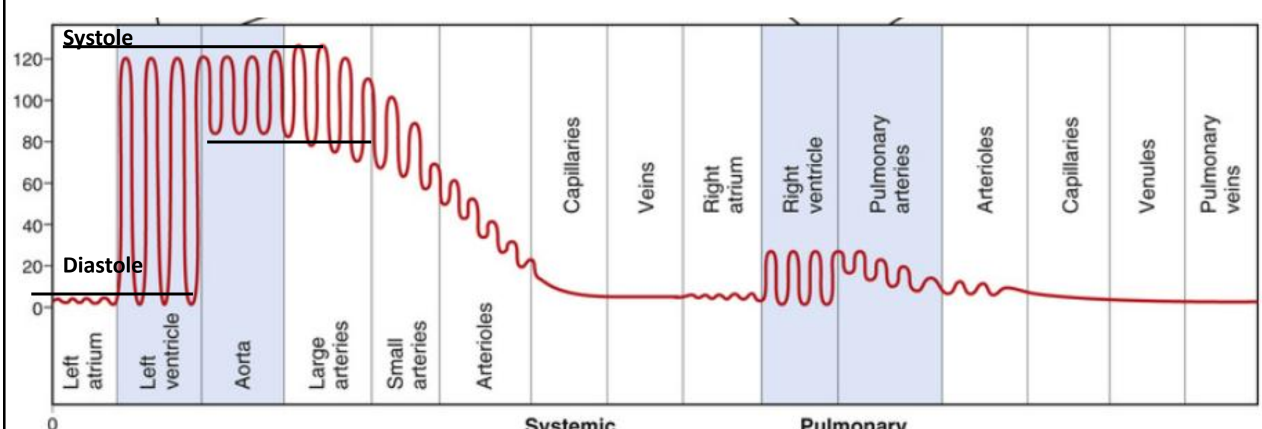
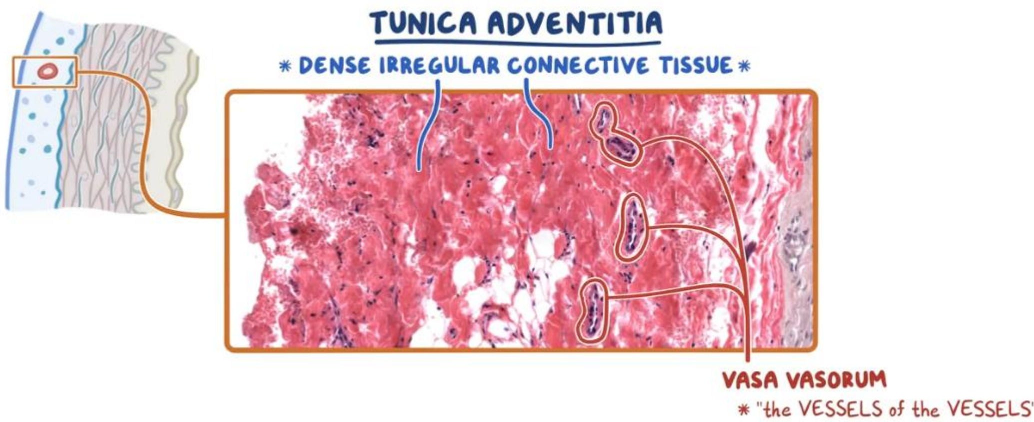


FIGURE 14-2 Normal blood pressures in the different portions of the circulatory system when a person is lying in the horizontal position.

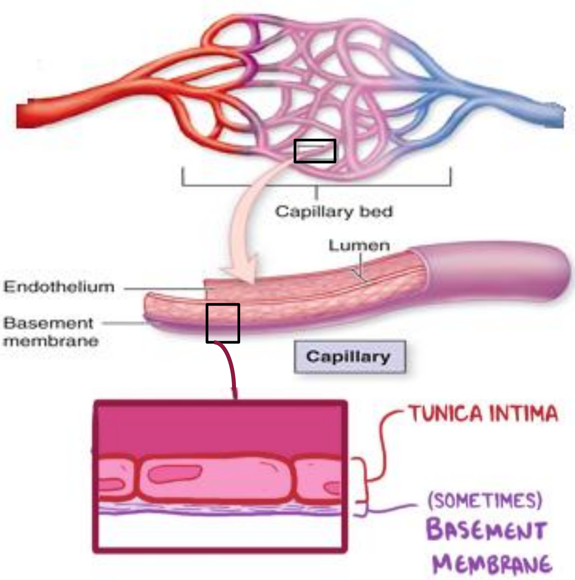
[VitalSource Bookshelf: Guyton and Hall Textbook of Medical Physiology E-Book]



3- Tunica adventitia (or externa): outmost layer of blood vessels. Composed of loose connective tissue with more collagen fibers to reinforce structure & nerve fibers. In large vessels, aorta and pulmonary artery, tunica adventitia contain blood vessels network named vasa vasorum to supply nutrition to the tunica externa.

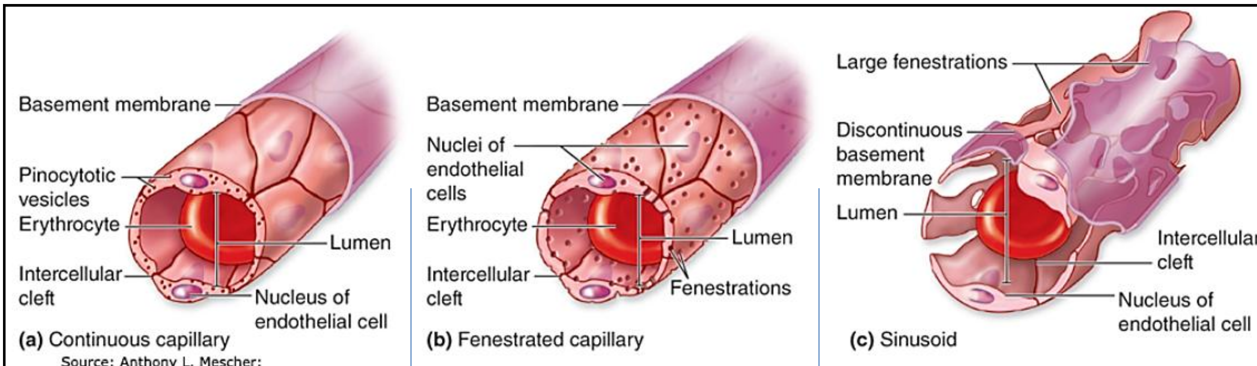
Capillaries

- Capillary bed is an interwoven network of capillaries which branch off the arterioles to perfuse tissues.
- Function: supply tissues with oxygen and nutrients, while removing carbon dioxide and waste products.
- Made up tunica intima or single layer of endothelial cells.



. Adapted from Mescher AL: Junqueira's Basic Histology Text and Atlas, 12th edition, McGraw-Hill, 2010.]

Picture imported from SEER Training (cancer.gov)



Tight occlusion junctions to yield minimal fluid leakage. Found in muscles, nervous system, lungs and exocrine glands

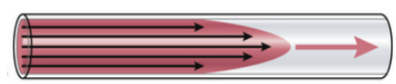
Small pores to allow the exchange of larger molecules. Found in kidney, choroid plexus and endocrine glands

Wide gaps between the endothelial cells. These capillaries allow the passage of white and red blood cells. Found in bone marrow, liver, and spleen.

Pressure, flow and resistance

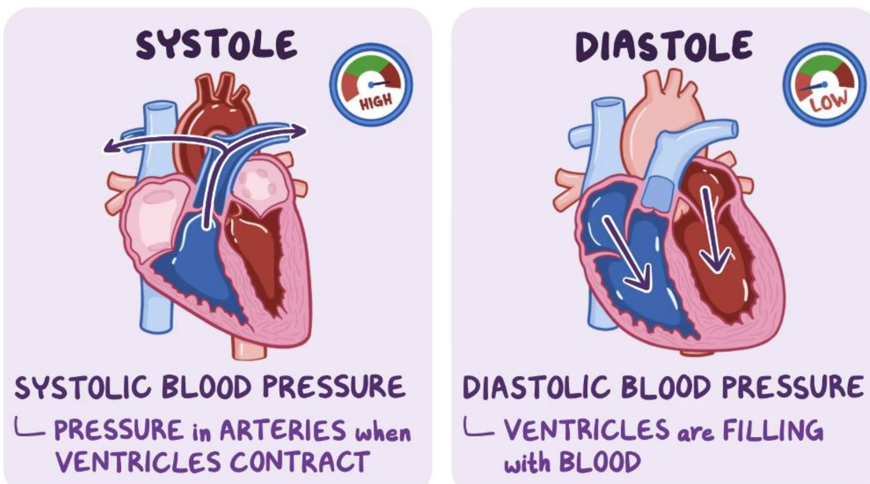
- Blood flow through a blood vessel is determined by two factors:
  - pressure gradient: pressure difference of blood between the two ends of the vessel, and the force that pushes the blood through the vessel.
  - vascular resistance: attributed to the friction between the flowing blood and the intravascular endothelium. Higher blood viscosity leads to a decreased blood flow due to increased vessel resistance (friction).

Blood flow



- The mechanical power of the heart as a pump is affected by the pressure (vascular resistance) and the volume of pumped blood.

Blood pressure



Normal BP at rest is <120 mmHg systolic and 80 mmHg diastolic

Mean arterial pressure

Mean arterial pressure: average pressure in the arteries in one cardiac cycle (systole and diastole). MAP is influenced by cardiac output and systemic vascular resistance.

Used to assess if blood flow through out the body is enough for perfusion.

- Normal value ranges from 70 -100 mmHg.
- Below 60 mmHg indicates that blood may not perfuse all organs (ischemia).
- Above 100 mmHg indicates excessive arterial pressure that can damage heart muscle

Factors affecting blood pressure:

1- Aging: causes an increase in systolic blood pressure because of decreased arterial compliance secondary to arteriosclerosis.

2- Activity level: for example, during sleep, blood pressure falls because of the body's decreased metabolic demands.

3- Anger and stress increase blood pressure, all because of sympathetically mediated responses.

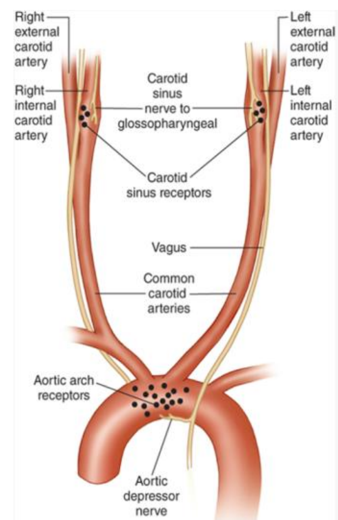
4-Pregnancy: Due to fluctuating hormones and changes in circulation, blood pressure can be below normal in the first and second trimesters.

Regulation of Blood Pressure

- Baroceptor reflex & MAP
- Vasoactive mediators
- Autonomic control via vasoconstriction and cardiac output
- Renin-angiotensin-aldosterone system

1- Baroreceptor reflex

- Baroreceptors are in the aortic arch and carotid sinus.
- Stretch receptors: respond to stretch or distension of the blood vessel walls.
- Decrease in mean arterial pressure generate action potentials (like pacemaker cells).
- These action potentials are transmitted to the vasomotor center in the medulla oblongata (cardio-regulatory).
- Baroreceptor reflex is a fast neural mechanism responsible for the minute-to-minute regulation of arterial blood pressure.



Negative feedback system

Increase of arterial pressure stimulates the baroreceptors and vasomotor center (via action potentials).

The vasomotor center adjusts ANS accordingly:

1- Stimulate the parasympathetic nervous system via vagal nerve. Also, Ach stimulate nitric oxide release from endothelial cells leading to vasodilation.

2- Decrease SNS input into the blood vessels and β1 receptors.

Therefore, stimulation of baroreceptor leads to reduced cardiac output, and dilates the blood vessels which brings the elevated blood pressure back to normal.

Baroceptor Dysfunction: Orthostatic hypotension

- Sudden drop in blood pressure upon standing from a sitting or supine position.
- Decrease in systolic blood pressure by ≥20 mm Hg or decrease of diastolic blood pressure by ≥10 mm Hg within 3 minutes of standing up. Most prevalent in patients aged ≥ 65 years.

- Etiology include baroreflex dysfunction, severe volume depletion, and adverse effects of medications.

- Symptoms can include dizziness, lightheadedness, and muscle ache in the neck and shoulders.

2- Vasoactive Mediators

Vasoactive substances are released from endothelial cells, endocrine glands, and cardiomyocytes. They affect vascular smooth muscle tone leading to vasoconstriction or vasodilation.

Vasoconstrictors include

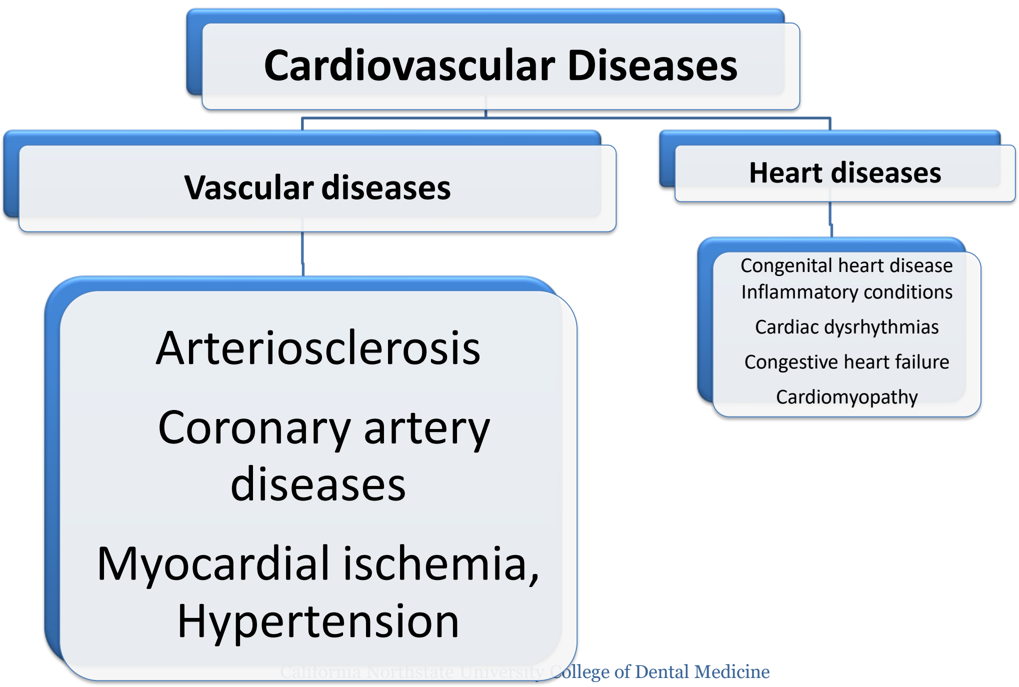
- **Catecholamines:** Epinephrin and norepinephrine from adrenal medulla in response to SNS stimulation to cause vasoconstriction.
- **Endothelin:** by vascular endothelium leading to vasoconstriction with positive inotropic and chronotropic effects on the heart.
- **Thromboxane A_2:** by vascular endothelium to cause vasoconstriction and stimulate platelet aggregation.

Vasodilators include:

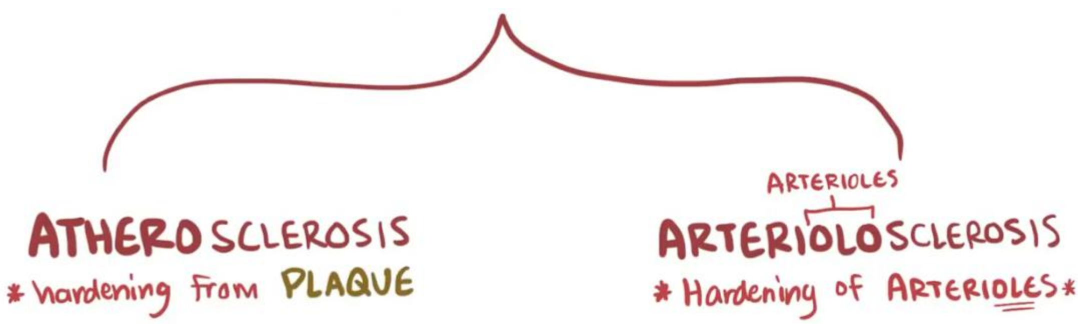
- **Nitric oxide (NO):** Most imp. vasodilator. Synthesized from arginine and oxygen via endothelial-derived nitric oxide synthase (eNOS) in response to acetylcholine, histamine, bradykinin, serotonin and others. Inhibition of NO synthesis results in a significant elevation of blood pressure.

- **Prostacyclin (PGI2):** synthesized by the endothelial cells. Opposite effects to TxA2. Prostacyclin causes vasodilation and inhibits platelet aggregation.

- **Atrial natriuretic peptide (ANP):** produced by specific cardiomyocytes in the heart atria. Secretion is stimulated by increased filling and stretch of the atria. ANP causes vasodilation, diuresis (increased urine production) leading to decreased blood volume and lowering blood pressure to normal levels.


Cardiovascular Diseases
Vascular diseases
Heart diseases
Arteriosclerosis
Coronary artery diseases
Myocardial ischemia, Hypertension
Congenital heart disease
Inflammatory conditions
Cardiac dysrhythmias
Congestive heart failure
Cardiomyopathy

Arteriosclerosis

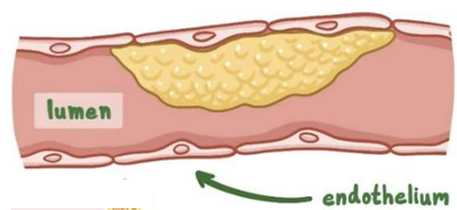


Arteriolosclerosis

- Hyaline arteriolosclerosis involves thickening of blood vessel wall in small arteries, arterioles.
- Vessel thickening causes a reduction in lumen size, increased rigidity and increased risk of vessel rupture.
- **Risk factors:** chronic hypertension and diabetes mellitus.
- **Manifestation:** Nephropathy (hyalinosis of renal arterioles leading to kidney damage) and increase risk of cerebral small vessel diseases (stroke and cognitive impairment).

Atherosclerosis

- More common.
- Atherosclerosis is a progressive, inflammatory disease the most commonly affect abdominal aorta, coronary, carotid arteries)



Pathogenesis: Inflammatory response to endothelial cell injury and deposition of atheromatous plaques (cholesterol deposits) within the tunica intima (arterial lumen) with subsequent inflammatory infiltration and partial occlusion of the artery.

Risk factors include hypertension, diabetes, smoking and dyslipidemia (increase in LDL or a decrease in HDL levels, why?) as well as the non-modifiable risk factors (age, family history, and African-American males).

Complications:

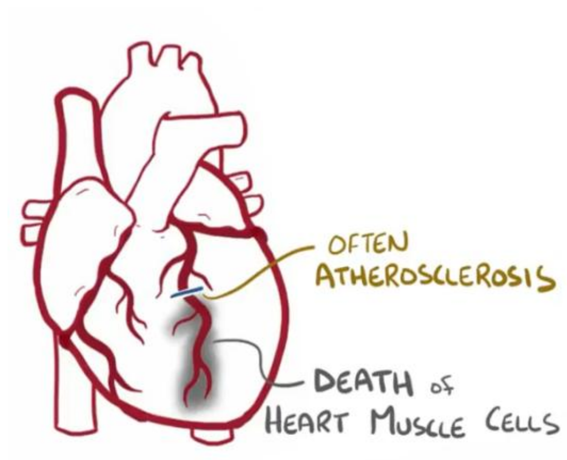
1- Ischemia to perfused organ (> 70 % occlusion of artery) leading to angina.

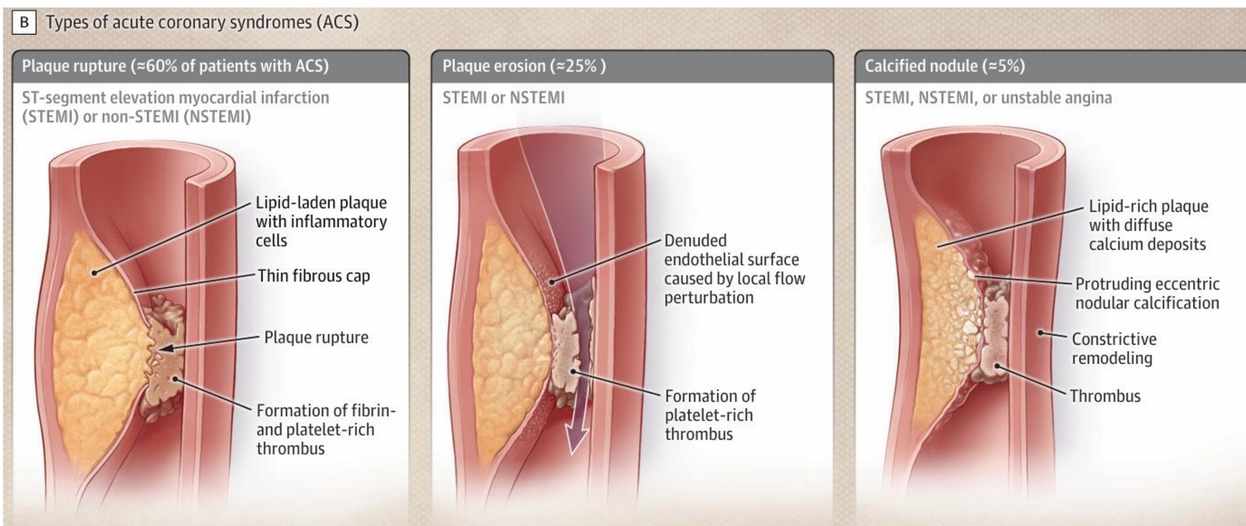
2- Plaque rupture causes accumulation of platelets and subsequent thrombosis leading to myocardial infarction, ischemic stroke.

3- Cholesterol emboli: part of cholesterol plaque break off and block other arteries. Complication of CVS procedure.

Myocardial infarction

- Coronary circulation supplies the heart cells supplied with oxygen and nutrients.
- Inadequate tissue perfusion = cardiomyocyte necrosis and MI. MI is acute & more severe than stable angina (irreversible cell injury vs reversible cell injury).





MI usually follows complete occlusion of a coronary artery (secondary to a raptured atherosclerotic plaque and thrombus formation).

[Diagnosis and Treatment of Acute Coronary Syndromes: A Review | Acute Coronary Syndromes | JAMA | JAMA Network]

Symptoms: can include compressing chest pain, chest tightness, diaphoresis, fatigue and dyspnea.

Diagnosis: ECG, blood parameter like cardiac-specific troponin and creatine kinase-MB fraction.

Complications: arrhythmias due to disruption of conductivity following damage. Cardiogenic shock (depending on damage). Pericarditis (due to inflammatory process). Tissue remodeling and heart failure.

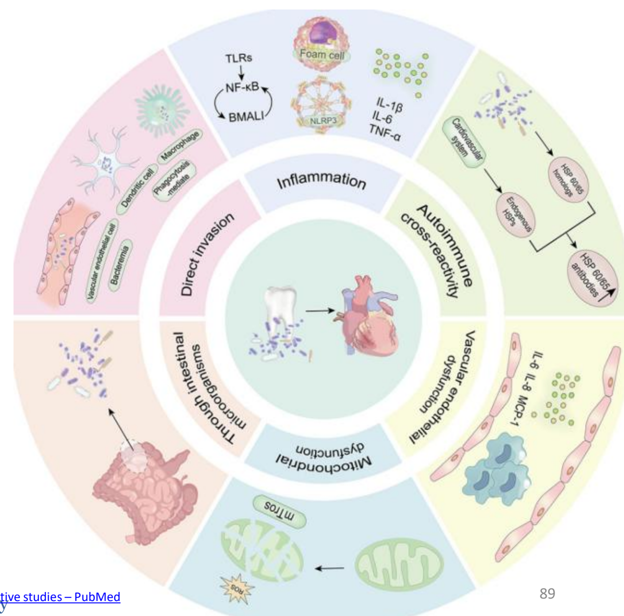
Management Guidelines

- **Pharmacological** Several including relieve pain, re-establish blood flow to relieve ischemia, minimize the size of the infarction among others.
- **Surgical**
- **Non-pharmacological** that include lifestyle or dietary changes.

Heart Disease and Stroke Statistics—2022 Update: A Report From the American Heart Association | Circulation (ahajournals.org)

Periodontal diseases & cardiovascular system

- Periodontitis is associated with increased risk of CVS disease including coronary heart disease, myocardial infarction, stroke, and increase cardiovascular mortality.
- Periodontitis can promote systemic inflammatory response, promote vascular injury and periodontal pathogens can cause vascular damage.
- Managing periodontal diseases can improve CVS outcomes/risk



[Periodontal disease and subsequent risk of cardiovascular outcome and all-cause mortality: A meta-analysis of prospective studies – PubMed]
[An update on the role and mechanisms of periodontitis in cardiovascular diseases - ScienceDirect]

California Northstate University

Recommended resources